MEDITERRANEAN CLIMATE

FIVE-REGION WORLD MAP → SEASONAL PRESSURE SWITCH → CLIMATE IDENTITY FIREWALL → LOCAL-WIND MATRIX → SCLEROPHYLL DESIGN → REGIONAL SCRUB NAMES

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

PRIMARY CORE FLOW • SUBORDINATE EXTRA • APPROVAL / REVIEW

Approval: FALSE • Pending user review • source generation g3 and all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles

00 — STAGE 00 FIVE-REGION WORLD MAP

FIVE-REGION WORLD MAP · MEDITERRANEAN BASIN · LARGEST EXPRESSION · CALIFORNIA + CENTRAL CHILE · WEST COASTS · SOUTH-WEST SOUTH AFRICA · SOUTH AUSTRALIA · FIFTH REGION

MEDITERRANEAN BASIN → largest expression

CALIFORNIA + CENTRAL CHILE → west coasts → CAPE → south-west South Africa → SW / SOUTH AUSTRALIA → fifth region

CORE CLAIM
- MEDITERRANEAN BASIN → largest expression
- CALIFORNIA + CENTRAL CHILE → west coasts

MECHANISM
- CAPE → south-west South Africa
- SW / SOUTH AUSTRALIA → fifth region

CONSEQUENCE / LIMIT
- MUST REMEMBER: Mediterranean Cs climate results from summer subtropical-high subsidence and winter westerly cyclone/frontal rain
- on western margins, supporting sclerophyll vegetation, winter cereals, vines, olives and high summer fire risk.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: MUST REMEMBER: Mediterranean Cs climate results from summer subtropical-high subsidence and winter westerly cyclone/frontal rain on western margins, supporting sclerophyll vegetation, winter cereals, vines, olives and high summer fire risk.

01 — STAGE 01 SEASONAL PRESSURE SWITCH

SEASONAL PRESSURE SWITCH · SUBTROPICAL HIGH SHIFTS POLEWARD · CLEAR SKY AND DROUGHT · WESTERLIES SHIFT EQUATORWARD · FRONTAL RAIN ARRIVES

CORE CLAIM
- SUMMER → subtropical high shifts poleward
- SUBSIDENCE → clear sky and drought

MECHANISM
- WINTER → westerlies shift equatorward
- DEPRESSIONS → frontal rain arrives

CONSEQUENCE / LIMIT
- SUMMER → subtropical high shifts poleward
- SUBSIDENCE → clear sky and drought

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: WINTER → westerlies shift equatorward; DEPRESSIONS → frontal rain arrives.

02 — STAGE 02 CLIMATE IDENTITY FIREWALL

CLIMATE IDENTITY FIREWALL · SUMMER DRY · WINTER WET · SUMMER WET · WINTER DRIER · EASTERN MARGIN · RAIN THROUGH MUCH OF YEAR · RAINFALL SEASON OUTRANKS LATITUDE ALONE

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
MEDITERRANEAN → summer dry / winter wet	EASTERN MARGIN → rain through much of year	MEDITERRANEAN → summer dry / winter wet
MONSOON → summer wet / winter drier	RULE → rainfall season outranks latitude alone	MONSOON → summer wet / winter drier

CORE CLAIM
- MEDITERRANEAN → summer dry / winter wet
- MONSOON → summer wet / winter drier

MECHANISM
- EASTERN MARGIN → rain through much of year
- RULE → rainfall season outranks latitude alone

CONSEQUENCE / LIMIT
- MEDITERRANEAN → summer dry / winter wet
- MONSOON → summer wet / winter drier

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: EASTERN MARGIN → rain through much of year.

03 — STAGE 03 LOCAL-WIND MATRIX

LOCAL-WIND MATRIX · HOT, DUSTY, SAHARA ORIGIN · COLD, RHONE-VALLEY OUTFLOW · COLD, ADRIATIC DOWNSLOPE WIND · LOCAL MEDITERRANEAN WINDS ARE NOT ALL HOT

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
SIROCCO → hot, dusty, Sahara origin	BORA → cold, Adriatic downslope wind	SIROCCO → hot, dusty, Sahara origin
MISTRAL → cold, Rhone-valley outflow	TRAP → local Mediterranean winds are not all hot	MISTRAL → cold, Rhone-valley outflow

CORE CLAIM **MECHANISM** **CONSEQUENCE / LIMIT**

- MONSOON → summer wet / winter drier
- RULE → rainfall season outranks latitude alone
- MONSOON → summer wet / winter drier

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: EASTERN MARGIN → rain through much of year.

03

STAGE 03

LOCAL-WIND MATRIX

LOCAL-WIND MATRIX · HOT, DUSTY, SAHARA ORIGIN · COLD, RHONE-VALLEY OUTFLOW · COLD, ADRIATIC DOWNSLOPE WIND · LOCAL MEDITERRANEAN WINDS ARE NOT ALL HOT

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
SIROCCO → hot, dusty, Sahara origin	BORA → cold, Adriatic downslope wind	SIROCCO → hot, dusty, Sahara origin
MISTRAL → cold, Rhone-valley outflow	TRAP → local Mediterranean winds are not all hot	MISTRAL → cold, Rhone-valley outflow

CORE CLAIM
- SIROCCO → hot, dusty, Sahara origin
- MISTRAL → cold, Rhone-valley outflow

MECHANISM
- BORA → cold, Adriatic downslope wind
- TRAP → local Mediterranean winds are not all hot

CONSEQUENCE / LIMIT
- SIROCCO → hot, dusty, Sahara origin
- MISTRAL → cold, Rhone-valley outflow

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: TRAP → local Mediterranean winds are not all hot.

04

STAGE 04

SCLEROPHYLL DESIGN

SCLEROPHYLL DESIGN · SMALL, HARD, WAXY OR LEATHERY · DEEP ACCESS TO STORED MOISTURE · DROUGHT AND FIRE PROTECTION · SCRUB OR OPEN WOODLAND

LEAF → small, hard, waxy or leathery

ROOT → deep access to stored moisture → BARK → drought and fire protection → STRUCTURE → scrub or open woodland

CORE CLAIM
- LEAF → small, hard, waxy or leathery
- ROOT → deep access to stored moisture

MECHANISM
- BARK → drought and fire protection
- STRUCTURE → scrub or open woodland

CONSEQUENCE / LIMIT
- LEAF → small, hard, waxy or leathery
- ROOT → deep access to stored moisture

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: BARK → drought and fire protection; STRUCTURE → scrub or open woodland.

05

STAGE 05

REGIONAL SCRUB NAMES

REGIONAL SCRUB NAMES · MEDITERRANEAN BASIN · FORM CONVERGES · SPECIES REMAIN REGIONAL

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
MAQUIS → Mediterranean basin	MALLEE → Australia	MAQUIS → Mediterranean basin
CHAPARRAL → California	FORM CONVERGES → species remain regional	CHAPARRAL → California

CORE CLAIM
- MAQUIS → Mediterranean basin
- CHAPARRAL → California

MECHANISM
- MALLEE → Australia
- FORM CONVERGES → species remain regional

CONSEQUENCE / LIMIT
- MAQUIS → Mediterranean basin
- CHAPARRAL → California

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: MALLEE → Australia; FORM CONVERGES → species remain regional.

06

STAGE 06

GROWTH-SEASON CALENDAR

GROWTH-SEASON CALENDAR · RAIN RETURNS AFTER DROUGHT · MOISTURE BUT COOLER GROWTH · BEST MOISTURE-TEMPERATURE OVERLAP · DROUGHT AND FIRE STRESS

AUTUMN → rain returns after drought

WINTER → moisture but cooler growth → SPRING → best moisture-temperature overlap → SUMMER → drought and fire stress

CORE CLAIM
- AUTUMN → rain returns after drought
- WINTER → moisture but cooler growth

MECHANISM
- SPRING → best moisture-temperature overlap
- SUMMER → drought and fire stress

CONSEQUENCE / LIMIT
- AUTUMN → rain returns after drought
- WINTER → moisture but cooler growth

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: MALLEE → Australia; FORM CONVERGES → species remain regional.

06

STAGE 06

GROWTH-SEASON CALENDAR

GROWTH-SEASON CALENDAR | RAIN RETURNS AFTER DROUGHT | MOISTURE BUT COOLER GROWTH | BEST MOISTURE-TEMPERATURE OVERLAP | DROUGHT AND FIRE STRESS

AUTUMN → rain returns after drought

WINTER → moisture but cooler growth → SPRING → best moisture-temperature overlap → SUMMER → drought and fire stress

CORE CLAIM

- AUTUMN → rain returns after drought
- WINTER → moisture but cooler growth

MECHANISM

- SPRING → best moisture-temperature overlap
- SUMMER → drought and fire stress

CONSEQUENCE / LIMIT

- AUTUMN → rain returns after drought
- WINTER → moisture but cooler growth

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: SPRING → best moisture-temperature overlap; SUMMER → drought and fire stress.

07

STAGE 07

AGRICULTURAL SYSTEM

AGRICULTURAL SYSTEM | WINTER RAIN | CEREALS AND SOIL RECHARGE | DEEP-ROOTED TREES | SURVIVE DRY SUMMER | SUN + IRRIGATION | FRUIT AND VEGETABLES | HIGH-VALUE EXPORT HORTICULTURE

WINTER RAIN → cereals and soil recharge → DEEP-ROOTED TREES → survive dry summer → SUN + IRRIGATION → fruit and vegetables → MARKETS → high-value export horticulture

CORE CLAIM

- WINTER RAIN → cereals and soil recharge
- DEEP-ROOTED TREES → survive dry summer

MECHANISM

- SUN + IRRIGATION → fruit and vegetables
- MARKETS → high-value export horticulture

CONSEQUENCE / LIMIT

- WINTER RAIN → cereals and soil recharge
- DEEP-ROOTED TREES → survive dry summer

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: SUN + IRRIGATION → fruit and vegetables; MARKETS → high-value export horticulture.

08

STAGE 08

WILDFIRE CHAIN

WILDFIRE CHAIN | WET SEASON | BIOMASS AND FINE FUEL | SUMMER DROUGHT | FUEL DESICCATION | HEAT + WIND + IGNITION | RAPID SPREAD | LAND MANAGEMENT

WET SEASON → biomass and fine fuel → SUMMER DROUGHT → fuel desiccation → HEAT + WIND + IGNITION → rapid spread → LAND MANAGEMENT → exposure and recovery

CORE CLAIM

- WET SEASON → biomass and fine fuel
- SUMMER DROUGHT → fuel desiccation

MECHANISM

- HEAT + WIND + IGNITION → rapid spread
- LAND MANAGEMENT → exposure and recovery

CONSEQUENCE / LIMIT

- WET SEASON → biomass and fine fuel
- SUMMER DROUGHT → fuel desiccation

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: HEAT + WIND + IGNITION → rapid spread.

09

STAGE 09

INDIA ANALOGUE MAP

INDIA ANALOGUE MAP | TEMPERATE FRUIT VALLEYS | KULLU AND SHIMLA APPLE BELTS | SUITABLE HILL ORCHARDS | GRAPE BELT, NOT CS CLIMATE | CLOSE DISTINCTION | MEDITERRANEAN CLIMATE IS NOT CONFINED TO THE MEDITERRANEAN BASIN

KASHMIR → temperate fruit valleys

HIMACHAL → Kullu and Shimla apple belts → UTTARAKHAND → suitable hill orchards → MAHARASHTRA → grape belt, not Cs climate

CORE CLAIM

- KASHMIR → temperate fruit valleys
- HIMACHAL → Kullu and Shimla apple belts

MECHANISM

- UTTARAKHAND → suitable hill orchards
- MAHARASHTRA → grape belt, not Cs climate

CONSEQUENCE / LIMIT

- CLOSE DISTINCTION: Mediterranean climate is not confined to the Mediterranean Basin, winter rainfall is not
- monsoonal reversal, and chaparral, maquis, fynbos and mallee are regional analogues rather than identical floras.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CLOSE DISTINCTION: Mediterranean climate is not confined to the Mediterranean Basin, winter rainfall is not monsoonal reversal, and chaparral, maquis, fynbos and mallee are regional analogues rather than identical floras.

KASHMIR → temperate fruit valleys

HIMACHAL → Kullu and Shimla apple belts → UTTARAKHAND → suitable hill orchards → MAHARASHTRA → grape belt, not Cs climate

CORE CLAIM
- KASHMIR → temperate fruit valleys
- HIMACHAL → Kullu and Shimla apple belts

MECHANISM
- UTTARAKHAND → suitable hill orchards
- MAHARASHTRA → grape belt, not Cs climate

CONSEQUENCE / LIMIT
- CLOSE DISTINCTION: Mediterranean climate is not confined to the Mediterranean Basin, winter rainfall is not
- monsoonal reversal, and chaparral, maquis, fynbos and mallee are regional analogues rather than identical floras.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CLOSE DISTINCTION: Mediterranean climate is not confined to the Mediterranean Basin, winter rainfall is not monsoonal reversal, and chaparral, maquis, fynbos and mallee are regional analogues rather than identical floras.

10

STAGE 10 APPLE VIABILITY LADDER

APPLE VIABILITY LADDER | WINTER CHILL | CULTIVAR-SPECIFIC DORMANCY | FROST, HAIL, ASPECT AND WATER | GRADING AND STORAGE | ROADS, COLD CHAIN AND TIMING | EVIDENCE LIMIT | FIRE IS CLIMATE-ENABLED BUT IGNITION, FUEL MANAGEMENT AND SETTLEMENT

WINTER CHILL → cultivar-specific dormancy

SITE → frost, hail, aspect and water → HARVEST → grading and storage → MARKET → roads, cold chain and timing

CORE CLAIM
- WINTER CHILL → cultivar-specific dormancy
- SITE → frost, hail, aspect and water
- HARVEST → grading and storage

MECHANISM
- MARKET → roads, cold chain and timing
- EVIDENCE LIMIT: Fire is climate-enabled but ignition, fuel management and settlement shape disaster
- Himalayan fruit belts share horticultural traits but not the complete

CONSEQUENCE / LIMIT
- Mediterranean circulation regime.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: EVIDENCE LIMIT: Fire is climate-enabled but ignition, fuel management and settlement shape disaster; Himalayan fruit belts share horticultural traits but not the complete.

11

STAGE 11 MEDITERRANEAN ANSWER SPINE

MEDITERRANEAN ANSWER SPINE | WESTERN MARGINS AT 30-45 DEGREES | SUMMER HIGH | WINTER WESTERLIES | SCLEROPHYLL, CROPS AND FIRE | INDIA IS AN ORCHARD ANALOGUE ONLY

LOCATE → western margins at 30-45 degrees → DERIVE → summer high / winter westerlies → LINK → sclerophyll, crops and fire → QUALIFY → India is an orchard analogue only

CORE CLAIM
- LOCATE → western margins at 30-45 degrees
- DERIVE → summer high / winter westerlies

MECHANISM
- LINK → sclerophyll, crops and fire
- QUALIFY → India is an orchard analogue only

CONSEQUENCE / LIMIT
- LOCATE → western margins at 30-45 degrees
- DERIVE → summer high / winter westerlies

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: QUALIFY → India is an orchard analogue only.

E

EXTRA SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT | INDIA HAS NO TRUE MEDITERRANEAN (CS) CLIMATE . | HIMALAYAN TEMPERATE HORTICULTURE IS INDIA'S MEDITERRANEAN-STYLE ORCHARD ANALOGUE. | HIMALAYAS ARE KNOWN FOR APPLE, PEACH, CHERRY, PEAR, MULBERRY | APPLE IS A TEMPERATE FRUIT CROP . | APPLE SUITABILITY DEPENDS ON WINTER CHILLING, FROST/HAIL EXPOSURE, GROWING-SEASON | NUMERICAL THRESHOLDS VARY BY CULTIVAR AND ORCHARD ALTITUDE; USE | APPLE ORCHARDS SHOULD BE FREE FROM HAILSTORMS AND FROST.

OPTIONAL SOURCE DEPTH
- India has no true Mediterranean (Cs) climate .
- Himalayan temperate horticulture is India's Mediterranean-style orchard analogue.
- Himalayas are known for apple, peach, cherry, pear, mulberry, walnut, almond and apricot orchards.
- Apple is a temperate fruit crop .

ADVANCED DEBATE
- Apple suitability depends on winter chilling, frost/hail exposure, growing-season temperature and water.
- Numerical thresholds vary by cultivar and orchard altitude; use them only with the source context.
- Apple orchards should be free from hailstorms and frost.
- Main apple areas: Kullu, Shimla, Kashmir Valley, hilly Uttarakhand .

LEGACY / NUANCE
- Apple is perishable and requires quick, efficient transport.
- Nashik and adjoining Maharashtra districts form a major grape/wine belt; Sangli is also important for grapes.
- NOT: India has a Mediterranean climate in Kashmir → India lacks true Cs; Kashmir is a temperate horticultural analogue.
- NOT: Indian grapes are grown mainly in the Himalayas → Main viticulture belt is Nashik–Sangli, Maharashtra.

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.

END OF MASTER CANVAS • poster embeds this exact master • tiled pages are overlapping crops of the same pixels